

Statistical Analysis of Alberta Wildfires from (2006 - 2024)

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Abstract

- maybe we want an abstract (which we will write at the end)?

Introduction

In Canada and across the world, we have seen an increase in forest area burned due to forest fires, with 2023 being an a record breaking year in forest damage for Canada (MacCarthy et al., 2025). Climate change is the main culprit, with changing weather patterns bringing heatwaves and drought which makes forests more easily ignitable than in the past (MacCarthy et al., 2025). Wildfires can be devastating to the forest ecosystem, but they also can have large ramifications on the economy and human health (Sadof & Anderson, 2023). When forests are burned, carbon dioxide is released into the air, further exacerbating our climate crisis and creating a positive feedback loop for future disastrous wildfires (Sadof & Anderson, 2023). Scientists have made significant efforts to understand forest fire dynamics allowing for improved forest fire mitigation and combative tactics *cite?*. However, it is of great importance to continue to research and understand how the forest fire landscape in Canada is changing so we can be better prepared to mitigate the environmental, economic, and social damages they will cause.

1. Purpose

Domain:

Our project looks at forest fire data in Alberta between 2006 and 2024. The data set gives information on total area burned, fire spread rate, causes of fires, as well as fire types, and time stamps and location coordinates Forestry & Parks (2025). We are narrowing our focus down to specifically focus on how weather factors, like wind speed, contribute to fire spread rate as well as how the cause of fires (specifically lightning) may be changing over time due to climate change.

Investigation Goal:

****Question 1:**** Research has indicated that the frequency of lightning caused forest fires is increasing due to climate change Beverly (2024). We want to see if the Alberta forest fire data corroborates this claim. We will compare the difference between the estimated proportion of lightning caused wildfires from 2024 to the estimated proportion of lightning caused wildfires from 2006 to see if the former is significantly larger than the latter.

****Question 2:**** There are a variety of environmental factors which affect how quickly a fire spreads, ultimately contributing to the overall impact and environmental burden that a forest fire has on its surrounding area. We are investigating how one key weather variable, wind speed, affects the fire spread rate.

Practical Implications: ~~#~~We can expand on this section

****Question 1**:** By examining how the proportion of lightning-caused wildfires has changed over time, we can assess one way in which climate change is reshaping wildfire patterns. In Canada, lightning already causes almost half of all wildfires and accounts for about 90% of burned forest area Coogan et al. (2025). Climate change is increasing both lightning frequency and the amount of dry, flammable debris in forests Public Health Agency of Canada (2024), suggesting more lightning-driven fires in the future. Examining these trends helps estimate climate impacts, inform on fossil fuel emission-reduction policies, and guide forest management practices such as prescribed burning Parks Canada (2025).

****Question 2**:** We are investigating how wind speed affects fire spread. In reality, wildfire models also include factors like humidity, temperature, topography, and fuel type (Natural Resources Canada, 2025). Understanding how wind accelerates the rate of fire spread helps predict fire risk to communities and habitats, guide evacuation needs, and assess environmental and economic impacts. High winds can also reduce the effectiveness of aerial fire suppression, making wind speed a key factor to understand in wildfire management Shmuel & Heifetz (2023).

Population(s) of Interest:

Our forest fire data set looks at wildfires in Alberta between 2006 and 2024. Specifically this data includes wildfires only occurring within the FPA (forest protection area), either provincial or federally owned/managed lands. For question 1, we will investigate forest fires in 2006 and 2024, with 1954 and 1230 wildfire entries, respectively. We will be comparing forest fires caused by lightning to those caused by all non-natural human activity. For question 2, we will study all forest fire entries that contain information for fire spread rate and wind speed. This accounts for around 89% $\frac{23657}{26551}$ of data entries.

Variables of Interest:

****Question 1**:** The variables we will analyze are fires within the “GENERAL_CAUSE” column of the data set which are “Lightning” (1 Success). All other fires within the “GENERAL_CAUSE” column which are not “Lightning” will be considered a 0 (Failure). We will analyze this variable for fires in the “YEAR” 2024 and compare it to fires from the “YEAR” 2006. ****Question 2**:** For the linear regression, our independent (x) variable will be the numerical variable “WIND_SPEED” (in km/h) and our dependant (y) variable will be the numerical variable “FIRE_SPREAD_RATE” (in m/min).

Data Collection Method:

****Quesetion 1**.** The “GENERAL_CAUSE” of wildfires is investigated by Alberta wildfire officials. Wildfires under investigation where the cause was not determined were labelled “Undetermined”. ***Question 2*.** “FIRE_SPREAD_RATE” and “WIND_SPEED” were recorded at the time of initial fire assessment. The exact assessors are Alberta Wildfire personnel and are indicated in the “ASSESSMENT_RESOURCE” column of the data set who assessed the fire at the recorded “ASSESSMENT_DATETIME”. Assessors include “air attack officer”, “initial action forces”, “wildfire assessor”, or “other”. More details on these assessment positions can be found in the Historical Wildfire Data Dictionary (Forestry and Parks, 2025).

2. Data

Source of Data:

This data comes from the Government of Alberta official website, published by Forestry and Parks. The full data set name is Historical wildfire data: 2006 to 2024.

Data Characteristics:

[Describe the nature of the data (e.g., categorical, continuous, time-series, text/unstructured) and the primary variables you will be using] New section

Permission/Accessibility:

This data set contains information licensed under the Open Government Licence - Alberta. This worldwide, royalty-free, perpetual, non-exclusive license allows us to copy, modify, publish, translate, adapt, distribute or otherwise use the information in any medium, mode or format for any lawful purpose Government of Alberta (n.d.).

Time Frame of Data Collection :

Data was collected from 2006-01-01 to 2024-12-31. The resource was created on 2020-01-09 and was last modified on 2025-01-31. Updates to this data set are irregular in frequency.

3. Topic(s) to Investigate

****Question 1**:** We want to see if the proportion of lightning caused wildfires from 2024 is significantly greater than the proportion of lightning caused wildfires from 2006. We hypothesize that 2024 will have a significantly higher proportion than wildfires from 2006 due to climate change which has resulted in a higher frequency of lightning strikes, and drier forest conditions beneficial for fire ignition upon a lightning strike. We will assess if $p_{lightning,2024} > p_{lightning,2006}$ as outlined by our hypotheses below.

$$H_0 : p_{lightning,2024} \leq p_{lightning,2006} \quad H_A : p_{lightning,2024} > p_{lightning,2006}$$

****Question 2**:** Wildfires are a persistent threat globally, with significant social, ecological, and economic impacts. Their spread and intensity result from complex interactions among fuels, topography, weather, and especially wind speed. This project will investigate the statistical relationship between wind speed and fire spread rate using historical wildfire data from 2006–2024. The primary research question asks: To what extent does wind speed predict the rate of wildfire spread? Understanding this relationship holds practical implications for fire behavior prediction, risk management, and the development of early warning systems for communities in wildfire-prone regions.

The population of interest includes all historical wildfires recorded within the provided dataset, with key variables being fire spread rate and wind speed. Both variables have been observed and recorded during wildfire events, with data carefully logged.

Wildfires are a recurring natural hazard in Alberta, Canada, where vast boreal forests, grasslands, and mixed-fuel landscapes are exposed to seasonal fire risk. Between 2006 and 2024, over 26,000 fire events were recorded in the Historical Wildfire Data data set (Forestry & Parks, 2025). This data set provides a unique opportunity to study the environmental drivers of fire behaviour in a consistent, long-term record. In Alberta, major fire events such as the 2016 Fort McMurray wildfire highlighted how fast-spreading fires can destroy thousands of structures, displace tens of thousands of people, and cost billions in recovery. Anticipating spread rate is therefore a central concern in wildfire management, where even small gains in prediction can inform life-saving interventions. Research shows that wildfire spread is influenced by a combination of fuels, weather, and topography (Canada, 2025). Among these, wind speed is consistently identified as one of the strongest drivers. Higher winds tilt flames forward, enhance radiant and convective heating of unburned fuels, and carry embers downwind, which accelerates the head fire and can ignite spot fires well ahead of the main front (Rothermel, 1972). Wind speed is also operationally relevant because it constrains firefighting tactics. Strong winds reduce the accuracy and effectiveness of aerial suppression, causing water and retardant drops to drift away from the target (Fogarty & Slijepcevic, 1997; Plucinski et al., 2025). They also increase risks to ground crews and limit the safe use of heavy equipment. Thus, understanding how spread responds to wind speed not only improves predictive science but also shapes suppression planning and resource allocation. In this project, we therefore hypothesize that wind speed will be a largely contributing factor to fire spread rate in Alberta. To test this, we will fit a simple linear regression model of the form: $\text{FIRE_SPREAD_RATE} \sim \text{WIND_SPEED}$

H_0 : There is no correlation between wind speed and fire spread rate.
 H_A : The alternative hypothesis is a correlation between wind speed and fire spread rate.

4. Statistical Methods

Planned Methods:

****Question 1**** : We will use a classical right sided Z test for proportions to determine if $p_{lightning,2024} > p_{lightning,2006}$ as determined by the formula:

$$Z_{obs} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{(p(1-p)(\frac{1}{n_1} + \frac{1}{n_2}))^{1/2}} \text{ where } p = \frac{x_1 + x_2}{n_1 + n_2} \quad \text{and} \quad p_1 - p_2 = 0 \text{ under the null hypothesis.}$$

We will reject the null hypothesis if $p < 0.05$. We have confirmed that the conditions to use this method are met. Our variables $p_{lightning,2024} > p_{lightning,2006}$ are from independent samples. Our n values are sufficiently large with $n_{2024} = 1230$ and $n_{2006} = 1954$ to use parametric tests relying on the central limit theorem. To confirm our methods, we will also use the conventional $\frac{+1}{+2}$ Agresti-coull confidence interval for the difference between proportions.

$$(\tilde{p}_1 - \tilde{p}_2) \pm z_{1-2/\alpha} \sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_1+2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_2+2}} \text{ where } \tilde{p}_1 = \frac{X_1+1}{n_1+2} \quad \text{and} \quad \tilde{p}_2 = \frac{X_2+1}{n_2+2}$$

We will compute this confidence interval for $p_{lightning,2024} - p_{lightning,2006}$ and if both bounds of the confidence interval are above 0, this will confirm that we should reject the null hypothesis and conclude that $p_{lightning,2024} > p_{lightning,2006}$. If the bounds encompass 0, we will conclude that there is no significant difference between $p_{lightning,2024}$ and $p_{lightning,2006}$.

****Question 2**** : Our aim is to estimate the strength and nature of the relationship between wind speed and fire spread rate using a simple linear regression.

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where:

- * i = fire spread rate for the i_{th} observation.
- * x_i = wind speed for the i_{th} observation.
- * β_0 = intercept (the estimated fire spread rate when wind speed is 0).
- * β_1 = slope (the expected change in fire spread rate for a one-unit increase in wind speed).
- * ϵ_i = random error term accounting for variation not explained by wind speed.

We will estimate β_0 and β_1 using least squares regression, which minimizes the sum of squared differences between observed fire spread rates and predicted values.

The null hypothesis is $H_0 : \beta_1 = 0$ The alternative hypothesis is $H_A : \beta_1 \neq 0$

We will then assess the strength of the relationship using: - The estimated slope β_1 - The coefficient of determination (R^2), which measures how much of the variation in fire spread rate is explained by wind speed. In the end, we will also utilize a residual plot to assess our regression.

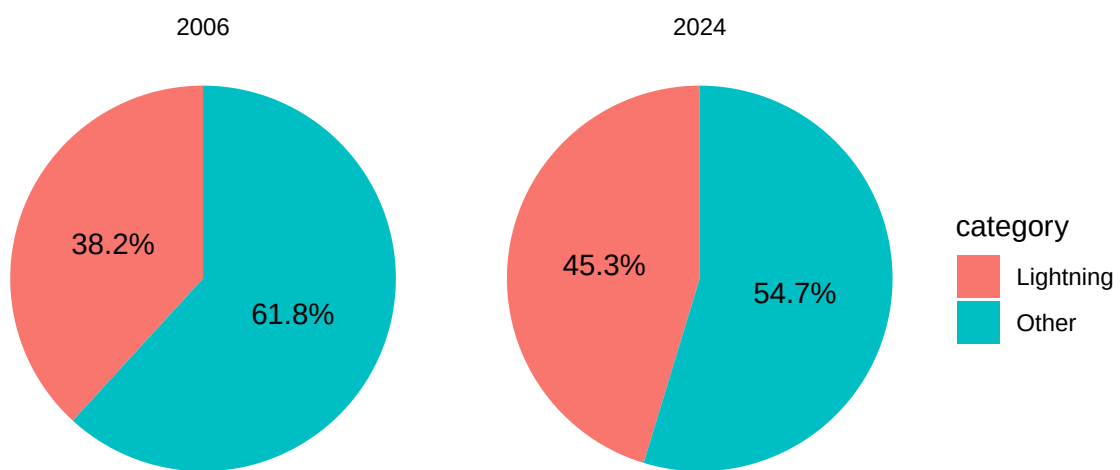


Figure 1: Proportion of Lightning-Caused Fires in 2006 and 2024

5. Results

We will investigate our initial question to see if the proportion of lightning-caused forest fires is significantly greater in 2024 than it is in 2006.

The initial sample proportions for each year are 0.38 and 0.45 for the year 2006 and 2024 respectively (Figure @ref(fig:piechart)).

Next we will confirm that our data follows the assumptions necessary to use a parametric test. (i.e. that n is sufficiently large under the central limit theorem).

Confirmed here, all values of n are sufficiently large to use the central limit theorem tests.

Testing hypothesis $H_0 : p_{2024} \leq p_{2006}$ and $H_A : p_{2024} > p_{2006}$

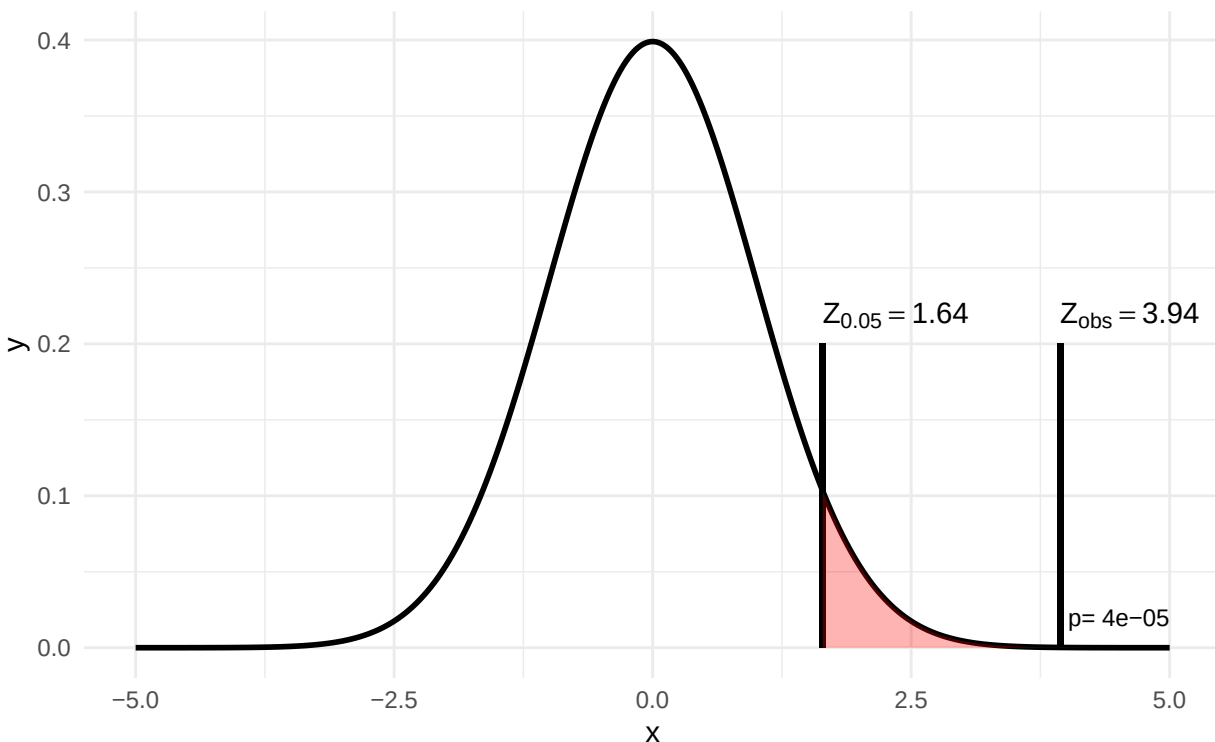
```
##### CALCULATING Z-observed #####
phat <- (x_2006 + x_2024)/(n_2006 + n_2024)
p1_min_p2 <- 0 #under the assumption of the null hypothesis
Zobs <- ((p_lightning_2024 - p_lightning_2006) - p1_min_p2)/sqrt(phat*(1-phat)*(1/n_2024 + 1/n_2006))
pvalue <- round(1 - pnorm(Zobs),5)
print(paste("The test statistic is:", round(Zobs,4), "The p-value is:", pvalue))
```

```
## [1] "The test statistic is: 3.9418 The p-value is: 4e-05"
```

Therefore, because our p-value is less than our alpha of 0.05, we can conclude that the proportion of lightning caused wildfires in 2024 is significantly greater than the proportion from 2006.

Right-Tailed Test for Lightning-Caused Wildfire Proportions $H_A: p_{2024} > p_{2006}$

Standard Normal Curve



Next, we will confirm our findings with a 95% Agresti-Coull Confidence interval for the difference in proportions, $p_{2024} - p_{2006}$.

```
##### USING THE AGRESTI-COULL (+1/+2 METHOD) 95% CI #####
p_tilde_2024 <- (x_2024 + 1)/(n_2024 + 2)
p_tilde_2006 <- (x_2006 + 1)/(n_2006 + 2)
```

```
ptilde_diff <- p_tilde_2024 - p_tilde_2006
z <- qnorm(1 - 0.05/2)
SE <- sqrt(((p_tilde_2024)*(1-p_tilde_2024)/(n_2024 + 2)) +
            ((p_tilde_2006)*(1-p_tilde_2006)/(n_2006 + 2))
            )
```

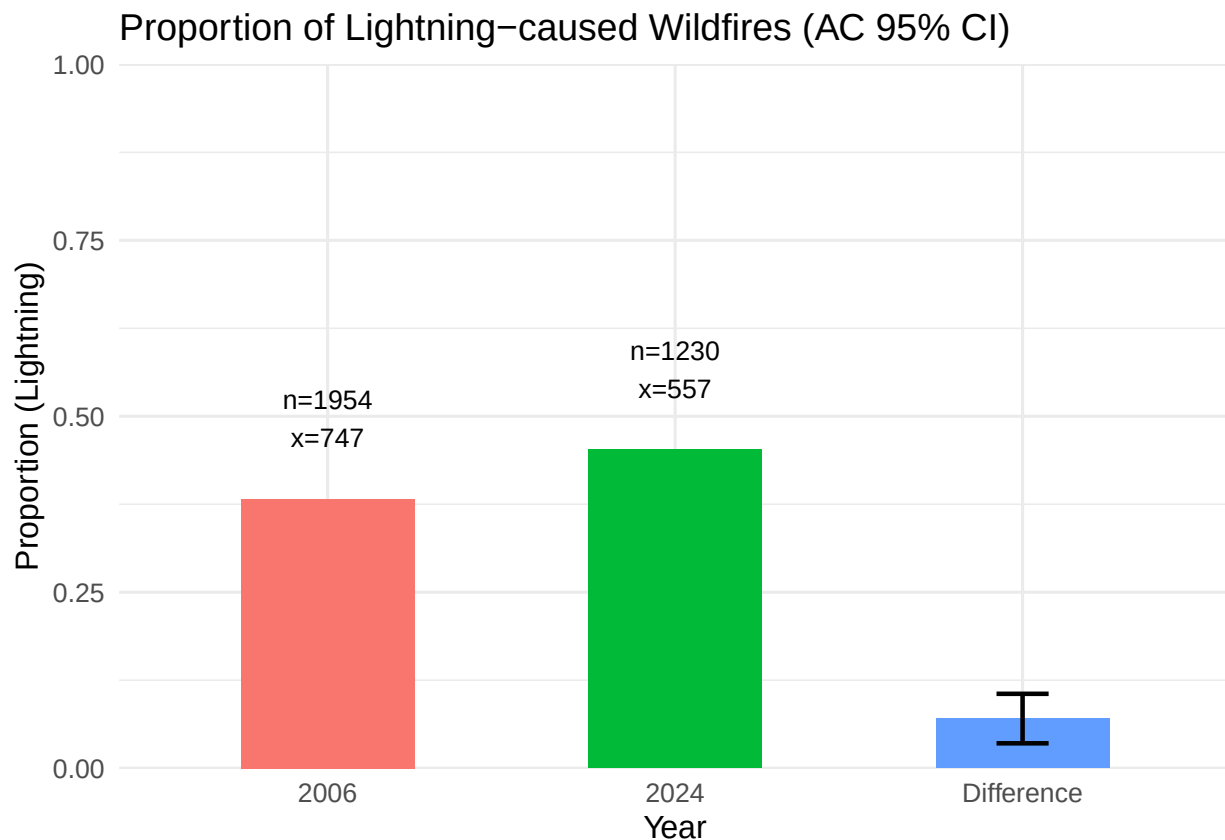
```
lb <- ptilde_diff - z * SE
ub <- ptilde_diff + z * SE
paste(round(lb, 4), "-", round(ub, 4))
```

```
## [1] "0.0353 - 0.1057"
```

```
# CONCLUSION : 2024 HAS A SIGNIFICANTLY HIGHER PROPORTION OF LIGHTNING-CAUSED WILDFIRES THAN 2006
```

Our 95% confidence interval for $p_{2024} - p_{2006}$ lies between 0.0353 - 0.1057. Both lower and upper bounds of our confidence interval are greater than 0, so this supports the above conclusion that the proportion of lightning caused wildfires from 2024 is significantly greater than that of 2006.

Here is a visualization for our 95% confidence interval.

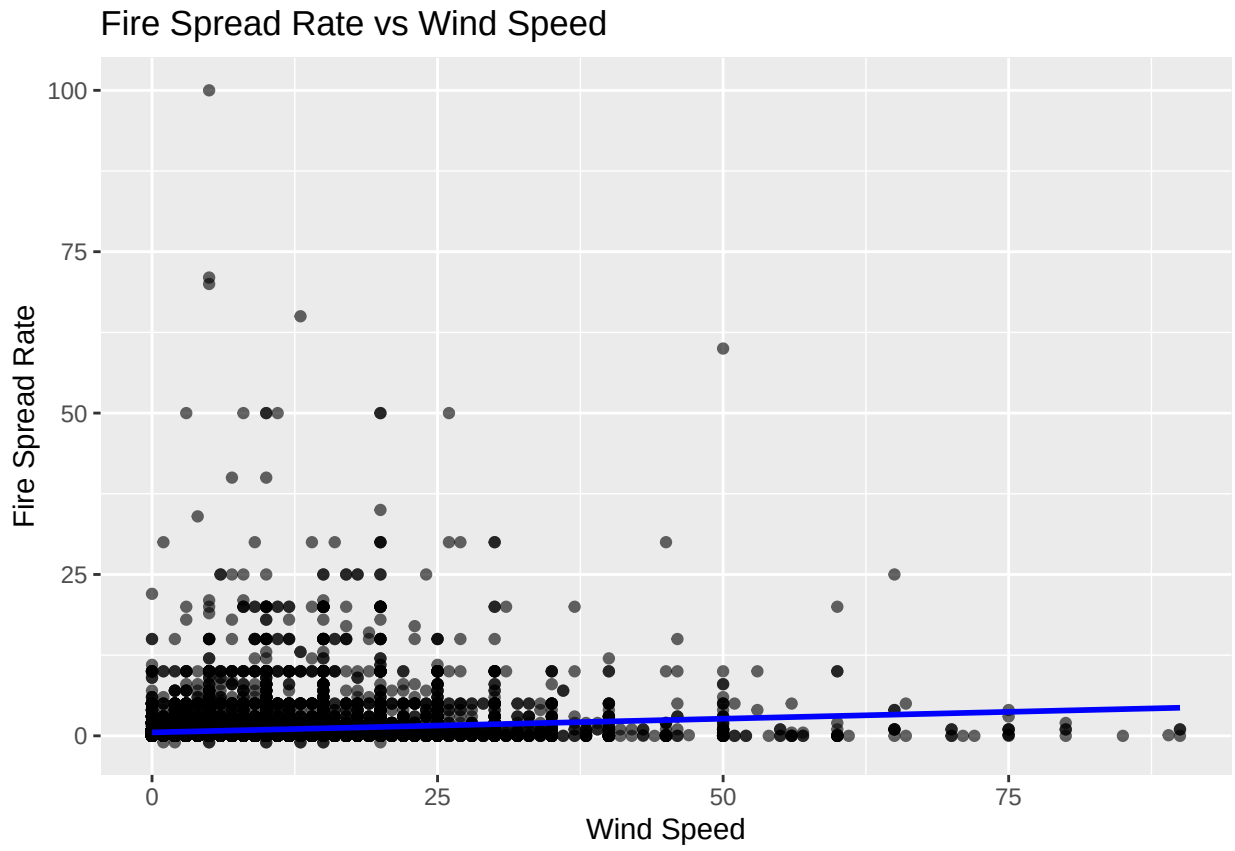


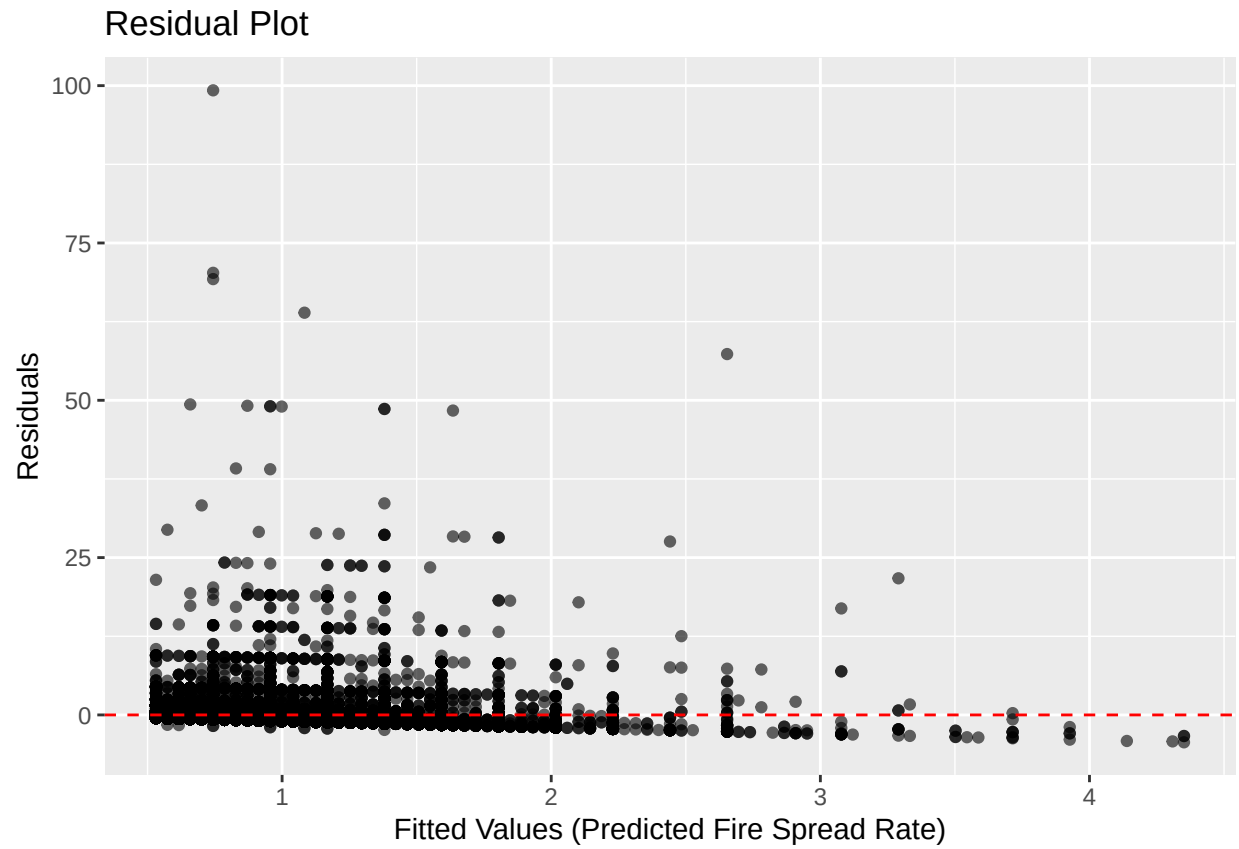
Question 2

```
# Create a clean dataset with only complete cases
wildfire_clean = df %>% filter(!is.na(WIND_SPEED) & !is.na(FIRE_SPREAD_RATE))
# Use the clean dataset
lm_model = lm(FIRE_SPREAD_RATE ~ WIND_SPEED, data = wildfire_clean)
summary(lm_model)
```

```
##
## Call:
## lm(formula = FIRE_SPREAD_RATE ~ WIND_SPEED, data = wildfire_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.351 -0.871 -0.601  0.044  99.256
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.531468   0.024340   21.84  <2e-16 ***
## WIND_SPEED    0.042441   0.001993   21.30  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.577 on 23655 degrees of freedom
## Multiple R-squared:  0.01882,    Adjusted R-squared:  0.01878
## F-statistic: 453.7 on 1 and 23655 DF,  p-value: < 2.2e-16

## 'geom_smooth()' using formula = 'y ~ x'
```

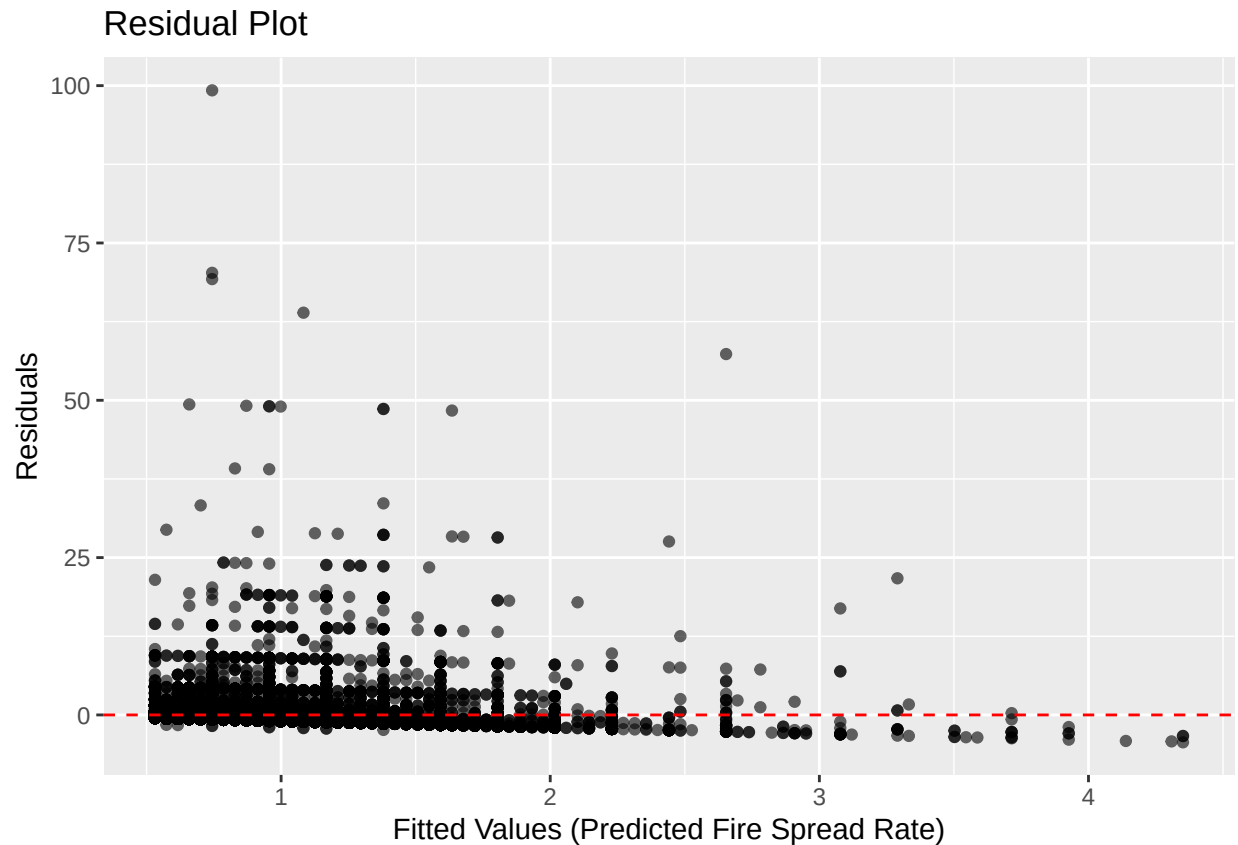




```
## Total observations in dataset: 26551
```

```
## Complete cases used in analysis: 23657
```

```
## Observations removed due to missing data: 2894
```



```
## Total observations in dataset: 26551

## Complete cases used in analysis: 23657

## Observations removed due to missing data: 2894
```

Conclusion And Future Work

6.1 Interpretation of Results [Discuss what the results imply in the context of the research questions and domain.] 6.2 Limitations [Highlight any limitations of the data or methodology that may impact conclusions.] 6.3 Recommendations and Future Directions [Suggest potential extensions of the study, alternative methods, or further investigations that could build on your findings.]

References

#Appendix

Here is some additional code that was not shown in the report

```
# Visualization for difference in proportions
plot_df <- tibble(
  YEAR = factor(c(2006, 2024, "Difference")),
```

```

prop = c(p_lightning_2006, p_lightning_2024, p_lightning_2024 - p_lightning_2006),
lower = c(NA, NA, lb),
upper = c(NA, NA, ub),
n      = c(n_2006, n_2024, NA),
x      = c(x_2006, x_2024, NA)
)

ggplot(plot_df, aes(x = YEAR, y = prop, fill = YEAR)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15, linewidth = 0.8) +
  geom_text(aes(label = ifelse(YEAR %in% c(2006, 2024), paste0("n=", n, "\n", x), NA)),
            vjust = -0.9, size = 3.5) +
  scale_y_continuous(limits = c(0, 1), expand = c(0, 0)) +
  labs(
    title = "Proportion of Lightning-caused Wildfires (AC 95% CI)",
    x = "Year",
    y = "Proportion (Lightning)"
  ) +
  theme_minimal(base_size = 12)

```

```

#Script for right-tailed test visualization
x <- seq(-5, 5, 0.001)
df_norm <- tibble(x = x, y = dnorm(x, mean = 0, sd = 1))

#labels for lines
labels_df <- tibble(x = c(qnorm(0.95), Zobs), y = 0.21,
                    label = c(bquote(Z[0.05] == .(round(qnorm(0.95), 2))), bquote(Z["obs"] == .(round(Zobs, 2)))))

p <- df_norm %>%
  ggplot(aes(x = x, y = y)) +
    geom_line(linewidth = 1) +
    geom_segment(aes(x = qnorm(0.95), xend = qnorm(0.95), y = 0, yend = 0.2),
                 linewidth = 1, color = "black") +
    geom_segment(aes(x = Zobs, xend = Zobs, y = 0, yend = 0.2),
                 linewidth = 1, color = "black") +
    geom_text(data = labels_df, aes(x = x, y = y, label = label), vjust = 0, hjust = 0, color = "black")
#shade in areas
geom_area(data = subset(df_norm, x > qnorm(0.95)), aes(x = x, y = y), fill = "red", alpha = 0.3) +
geom_area(data = subset(df_norm, x > Zobs), aes(x = x, y = y), fill = "blue", alpha = 1) + #shade
annotate(geom = "text", x = 5, y = 0.02, label = paste("p=", pvalue), size = 3, hjust = 1) +
ggtitle(bquote("Right-Tailed Test for Lightning-Caused Wildfire Proportions H" [A] * ": p" [2024] > "p" [2006]))
  subtitle = "Standard Normal Curve" +
  theme_minimal()
print(p)

```

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